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Prof. Kenneth M. Merz, Jr., Editor-in-Chief
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Dear Prof. Merz,

We submit our manuscript **"Resistance-Resilient Antimalarial Leads from African Natural Products: MD Validation Against Resistance Mutants"** for consideration as an Article in *JCIM*.

This work presents the first systematic MD validation of antimalarial leads against a panel of African-prevalent resistance mutations, with production MD (40 ns across 4 wild-type complexes) and docking-based scoring across 220 protein–ligand systems. We validate 20 candidates from a generative antimalarial library against six clinically relevant mutations (PfDHFR-N51I, C59R, S108N, I164L; PfCRT-K76T, K76A) using MM-GBSA free energy calculations. Three novel metrics are introduced: the Resistance Resilience Score (RRS) classifying compounds by binding resilience across mutants, the African Chemical Space Index (ACSI) quantifying NP character retention (23.5 % highly NP-like), and the Polypharmacology Network Score (PNS) weighting target engagement by STRING interactome centrality. External Tartarus benchmarking confirms that MPO drug-likeness and binding affinity are orthogonal (Spearman $\rho \approx 0$, $p = 0.09$).

All protocols, trajectories, and scoring outputs are deposited on Zenodo and GitHub under open-source licences. Statistical robustness is ensured through bootstrap confidence intervals and Bonferroni-corrected hypothesis testing.

Suggested reviewers.

1. Dr. Fidele Ntie-Kang, University of Buea — African NP computational discovery
2. Dr. Kelly Chibale, University of Cape Town — antimalarial drug discovery
3. Dr. John D. Chodera, Memorial Sloan Kettering — free energy methods, MD best practices

This manuscript is original, not under consideration elsewhere, and all authors have approved the submission.

Sincerely,

Myke-Vital Temgoua Sao
(on behalf of all co-authors)